

CURAM-AI

Phase 1 – Feasibility Sprint Report

Barrett Engineering Pty Ltd

Engineering & Construction Sector Analysis

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Trial Code: P1-2026-SAMPLE

1. Executive Summary

This feasibility sprint tested 15 documents from Barrett Engineering Pty Ltd to validate AI-powered document extraction accuracy and straight-through processing rates.

Metric	Result	Threshold	Status
Field Accuracy	96.2%	≥ 90%	PASS
STP Rate	92.5%	≥ 60%	PASS
Exceptions	11	< 8	REVIEW

Indicative Value Assessment

Tier 1: Time Savings	Tier 2: Error Reduction	Total Annual Value
\$79,040	\$983	\$80,023

Recommendation: All tests passed. Proceed to Phase 2 deep-dive analysis.

2. Technical Validation

2.1 Test Methodology

The feasibility sprint tested 15 documents provided by the client, distributed across 5 document categories. Each document was processed through the Curam-Ai extraction pipeline using Google Gemini 2.5 Flash for text extraction and structured data parsing.

2.2 Accuracy Criteria

- Pass Threshold: $\geq 90\%$ field-level accuracy
- STP Threshold: $\geq 60\%$ straight-through processing rate
- Exception Threshold: < 8 documents requiring manual review

2.3 Processing Results

Total Fields Extracted: 531

Correct Extractions: 511 (96.2%)

Flagged for Review: 20

False Positives: 3

3. Accuracy by Document Category

Category	Documents	Fields	Correct	Accuracy	Status
Vendor Invoice	5	109	104	95.4%	PASS
Beam Schedule	3	124	118	95.2%	PASS
Column Schedule	2	100	97	97.0%	PASS
Transmittal	3	78	75	96.2%	PASS
Drawing Register	2	120	117	97.5%	PASS

3.1 Edge Cases Identified

3 documents were flagged as edge cases requiring special handling:

- **INV-2024-003_HandwrittenPO.pdf** (Vendor Invoice) - Handwritten annotations difficult to parse
- **Transmittal_2024-157_Scanned.pdf** (Transmittal) - Low-quality scan, coffee stain on header
- **BeamSchedule_L03_Legacy.pdf** (Beam Schedule) - Old CAD format, non-standard table layout

4. Value Assessment

4.1 Business Profile

Parameter	Value	Notes
Total Staff	45	Firm headcount
Documentation Staff	28	Staff handling documents
Blended Hourly Rate	\$95	Including on-costs
Weekly Document Volume	150	Documents per week
Annual Volume	7,800	Projected annual throughput
Manual Processing Time	8 min	Per document
Current Error Rate	3.5%	Estimated baseline
Target STP Rate	80%	Automation target

4.2 ROI Calculation

Tier 1: Time Savings from Straight-Through Processing

Annual documents (7,800) × Manual time (8 min) × Hourly rate (\$95/hr) × STP rate (80%)
= **\$79,040** annual savings

Tier 2: Error Reduction Savings

Annual documents (7,800) × Error rate (3.5%) × Error reduction (3%) × Correction cost (\$120)
= **\$983** annual savings

Total Indicative Annual Value: \$80,023

5. Limitations & Assumptions

1. Accuracy metrics based on 15 sample documents; production accuracy may vary with document diversity.
2. ROI calculations assume 150 documents/week at current volume; actual savings scale with volume changes.
3. Blended hourly rate of \$95/hr includes on-costs; validate this aligns with internal cost models.
4. Error reduction projections assume 3% of flagged documents prevent costly downstream errors; adjust based on actual error impact.
5. Target STP rate of 80% is achievable with current document quality; may require improvement for edge cases.
6. Some legacy drawings in TIFF format may require additional preprocessing.

6. Phase 2 Discovery Questions

The following questions will be addressed in the Phase 2 deep-dive analysis to develop a board-ready business case and implementation roadmap:

1. How should the system handle the 3 edge-case document format(s) identified?
2. Should we leverage your Microsoft 365 environment for Power Automate integration and SharePoint document storage?
3. What ERP/accounting systems would require direct API integration for extracted data?
4. Are there approval workflows that should be built into the exception handling process?
5. What security/compliance requirements (ISO 27001, SOC 2, etc.) need to be addressed in the architecture?

7. Conclusion & Next Steps

Based on the feasibility sprint results, **Barrett Engineering Pty Ltd** has achieved the required accuracy thresholds for AI-powered document automation. With 96.2% field accuracy and 92.5% straight-through processing rate, the technology is ready for Phase 2 implementation planning.

The indicative annual value of **\$80,023** represents conservative estimates based on the business profile provided. Phase 2 will refine these calculations with detailed process mapping and integration requirements.

Next Steps:

- Schedule Phase 2 discovery call to review detailed requirements
- Provide additional sample documents for any new document types
- Identify key stakeholders for the implementation project
- Review integration requirements with IT/operations teams

For questions about this report:

Curam-Ai Protocol

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